Stochastic Dynamics of a Quantum Oscillator under Parity Measurement

Marina Benítez Sánchez, Luísa Toledo Tude
Instituto de Física Interdisciplinar y Sistemas Complejos, IFISC (CSIC-UIB)
Campus Universitat de les Illes Balears, E-07122 Palma de Mallorca, Spain

Abstract

Quantum optics explores the intrinsically quantum properties of light, where coherent states provide a fundamental bridge between classical and quantum behavior. These states are key resources for quantum technology and information processing. In this context, the control and understanding of how to extract information from such states becomes fundamental. In this work, we investigate how different schemes used for measuring the parity of a single quantized cavity mode under coherent displacements can alter its dynamics. Specifically, we analyze three distinct physical measurement scenarios: projective parity measurements at fixed intervals, stochastic measurements at random times, and their continuous ensemble-average limit, given by the Lindblad equation. Our results illustrate how measurement can be used not only to extract information, but also to manipulate the dynamics of a system.

1 Introduction

At the beginning of the twentieth century, experimental observations such as blackbody radiation and the photoelectric effect revealed the limitations of classical physics and led to the development of quantum theory. In 1900, Max Planck postulated that the energy of the light field was quantised [1], and the principle of wave–particle duality were introduced into optics by the Einstein equation, $E = h\nu$, which relates the energy E of the light corpuscle (photon), the frequency ν of the associated electromagnetic wave, and Planck's constant h [2].

The invention of the laser in 1960 provided a highly coherent source of light and contributed to the development of quantum optics. In this framework, coherent states, originally introduced by Schrödinger, became particularly important because they provide a natural connection between the classical and quantum descriptions of light. Their properties make them fundamental in quantum optics and relevant to modern quantum technologies, including quantum information processing [1].

A major challenge in quantum systems is maintaining and controlling their quantum properties in the presence of decoherence and environmental interactions. Moreover, quantum measurements are not passive observations: extracting information from a system modifies its state through measurement back-action. In cavity quantum electrodynamics (cavity QED), where an electromagnetic field is confined inside a highly reflective cavity, quantum states can be prepared, manipulated, and monitored with high precision. This makes cavity QED a suitable platform for studying measurement-induced dynamics and for exploring measurement back-action as a resource rather than simply as a source of disturbance.

An especially interesting measurement in this context is photon-number parity, which determines whether the number of photons in the cavity is even or odd. Parity measurements can be used to characterize non-classical states [4] and, when applied to a coherent state, can project the field onto an even- or odd-parity superposition of coherent states, known as a Schrödinger cat state [5]. These non-Gaussian states are of particular interest in quantum information processing and quantum error correction [4].

When parity measurements are performed continuously, the measurement back-action becomes part of the system's dynamics rather than an isolated operation. The resulting interplay between coherent evolution, dissipation, and continuous measurement can generate complex nonlinear dynamics. This motivates the use of continuous quantum measurements not only to extract information from a system, but also to engineer its evolution.

All in all, the aim of this work is to investigate how different measurement schemes affect the dynamics of a cavity QED system. We consider three physical scenarios: projective parity measurements performed at fixed time intervals, with a displacement of amplitude α between measurements (section 3.1); stochastic parity measurements occurring at random times, which can represent either measurement uncertainty or random interactions with the environment (section 3.2); and the ensemble-averaged dynamics obtained from many stochastic trajectories, corresponding to continuous interaction with the environment (section 3.2).

2 Theoretical concepts

2.1 Quantized electromagnetic field in a cavity

To understand the physical system considered in this work, it is useful to start from the classical description of an electromagnetic field confined inside a cavity. The cavity imposes boundary conditions on the electromagnetic field, so that not every spatial field configuration is allowed. Instead, the field can be decomposed into a discrete set of normal modes, each characterized by a particular spatial distribution and oscillation frequency ω_κ. These modes provide the natural degrees of freedom of the electromagnetic field inside the cavity [3].

For an ideal cavity, the energy of the electromagnetic field can be written as a sum of independent contributions from the different modes. Each contribution has the mathematical form of the energy of a classical harmonic oscillator [2],

$$H_{\mathrm{EM}} = \sum_{\kappa} \frac{1}{2} \left(p_\kappa^2 + \omega_\kappa^2 q_\kappa^2 \right), \tag{2.1}$$

where q_κ and p_κ are the classical canonical variables associated with the κ-th electromagnetic mode. Thus, although the electromagnetic field is a spatially extended physical object, its dynamics can be reduced to a collection of independent harmonic oscillators, one for each allowed cavity mode.

In the present work, we consider only one of these modes, with frequency ω_c. This single-mode approximation allows the electromagnetic field to be described by the dynamics of a single harmonic oscillator. It is this single cavity mode that will constitute the quantum system studied throughout the rest of the work.

Quantization is performed by promoting these classical variables to Hermitian operators $\hat{q}$ and $\hat{p}$ satisfying the canonical commutation relation

$$[\hat{q}, \hat{p}] = i\hbar. \tag{2.2}$$

The Hamiltonian of the cavity mode then becomes

$$\hat{H}_{\mathrm{cav}} = \frac{1}{2} \left(\hat{p}^2 + \omega_c^2 \hat{q}^2 \right). \tag{2.3}$$

which is a mathematical description of the energy dynamics of one electromagnetic mode. The two canonical variables $\hat{q}$ and $\hat{p}$ describe the two conjugate degrees of freedom of that mode, in the same way that position and momentum describe a mechanical harmonic oscillator [8].

It is convenient to introduce the annihilation and creation operators

$$\hat{a} = \sqrt{\frac{\omega_c}{2\hbar}}\, \hat{q} + \frac{i}{\sqrt{2\hbar\omega_c}}\, \hat{p}, \tag{2.4}$$

and

$$\hat{a}^\dagger = \sqrt{\frac{\omega_c}{2\hbar}}\, \hat{q} - \frac{i}{\sqrt{2\hbar\omega_c}}\, \hat{p}, \tag{2.5}$$

which satisfy

$$[\hat{a}, \hat{a}^\dagger] = 1. \tag{2.6}$$

In terms of these operators, the Hamiltonian takes the form

$$\hat{H}_{\mathrm{cav}} = \hbar\omega_c \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right). \tag{2.7}$$

The operator $\hat{n} = \hat{a}^\dagger \hat{a}$ counts the number of excitations of the selected cavity mode. The elementary excitation of the quantized electromagnetic field is a photon, so that $|n\rangle$ describes

a state with n photons occupying the same cavity mode, that is why it is called the photon-number operator. Consequently, the energy of the mode is quantized: its possible energy values are separated by integer and the energy eigenvalues of the cavity mode are

$$E_n = \hbar\omega_c \left(n + \frac{1}{2}\right). \tag{2.8}$$

Its eigenstates form the Fock basis,

$$\hat{n}\,|n\rangle = n\,|n\rangle\,. \qquad n = 0, 1, 2, \ldots \tag{2.9}$$

The spatial wavefunctions corresponding to the first eight of these states are illustrated in Fig. 1, displaying the characteristic spatial distributions, parity, and nodal structures associated with different excitation numbers.

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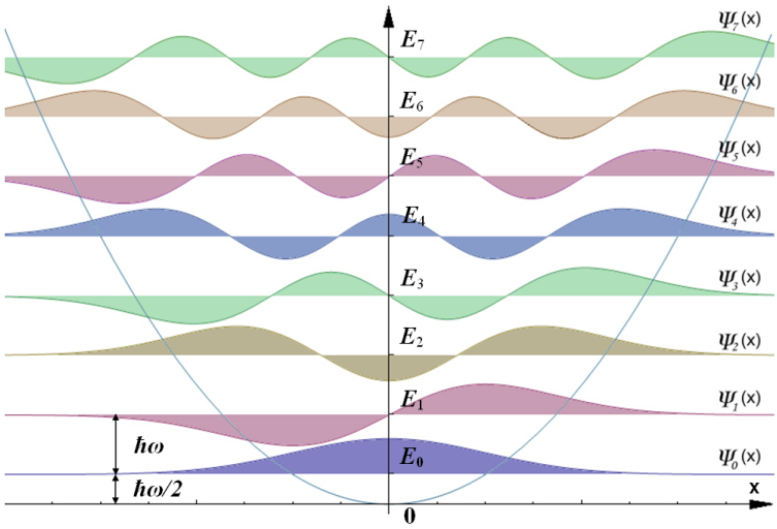


Figure 1: Wave functions of the first eight eigenstates of the quantum harmonic oscillator, corresponding to $n = 0, \ldots, 7$. The horizontal axis represents the position in units of $\sqrt{\hbar/(m\omega)}$ [6].

Furthermore, the creation and annihilation operators act on these states as

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,. \tag{2.10}$$

In particular, the vacuum state $|0\rangle$ is such that

$$\hat{a}\,|0\rangle = 0, \tag{2.11}$$

and, starting from the vacuum, all Fock states can be generated by repeatedly applying the creation operator,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle\,. \tag{2.12}$$

This quantization gives a direct physical interpretation to the Fock states: each excitation of the cavity mode contributes an energy $\hbar\omega_c$. In this sense, the cavity mode specifies the allowed electromagnetic field configuration, while the photon number specifies the number of quantized excitations of that mode.

The Fock basis will be particularly important for the present work because photon-number parity, introduced in the following section, is directly determined by whether the photon number is even or odd.

2.2 Coherent states and phase-space representation

Although Fock states provide a natural basis for the quantized electromagnetic field, they are not the only useful description of a cavity state. In particular, coherent states provide a connection between the quantum description of the field and the classical electromagnetic field. Indeed, they are the quantum states that most closely resemble classical electromagnetic fields.

A coherent state $|\alpha\rangle$ is defined as an eigenstate of the annihilation operator [8],

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle\,, \tag{2.13}$$

where α is a complex number.

Expanding the coherent state in the Fock basis gives

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,. \tag{2.14}$$

Consequently, the probability of finding n photons in a coherent state follows a Poisson distribution

$$P(n) = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2}\,\frac{|\alpha|^{2n}}{n!}. \tag{2.15}$$

with mean photon number

$$\langle \hat{n}\rangle = \langle\alpha|\hat{n}|\alpha\rangle = |\alpha|^2. \tag{2.16}$$

Coherent states can be generated from the vacuum by means of the displacement operator [7],

$$\hat{D}(\alpha) = \exp\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right). \tag{2.17}$$

Acting on the vacuum state, it produces

$$|\alpha\rangle = \hat{D}(\alpha)\,|0\rangle\,. \tag{2.18}$$

The displacement operator changes the phase-space position of a state. In particular,

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha. \tag{2.19}$$

Therefore, applying $\hat{D}(\alpha)$ to a coherent state results in

$$\hat{D}(\beta)\,|\alpha\rangle = e^{i\phi}\,|\alpha+\beta\rangle\,, \tag{2.20}$$

where the phase ϕ is physically irrelevant for the state representation. Thus, a displacement translates the state in phase space without changing its shape.

This operation is a central element of the protocol investigated in this work. In the ideal projective-measurement setting, the displacement operation is applied between successive parity measurements, producing a measurement-driven evolution of the cavity state.

It should be noted that the complex parameter α provides a natural connection between the quantum state of the cavity and the classical phase space of a harmonic oscillator. Writing

$$\alpha = X + iP, \tag{2.21}$$

the real and imaginary parts of α can be associated with two dimensionless field quadratures. Consequently, a coherent state can be represented as a localized distribution around a point (X, P) in phase space.

To represent quantum states in this phase-space picture, one can introduce the Wigner function [3], $W(\alpha)$, which is a quasiprobability distribution defined over the phase space of the cavity mode. For a density operator ρ, it is defined as

$$W(x,p) = \int_{-\infty}^{\infty} \langle x+y|\;\rho\,|x-y\rangle\, e^{2ipy}\,dy \tag{2.22}$$

where $|x\rangle$ denotes an eigenstate of the quadrature operator X. Unlike a classical probability distribution, it is not required to be positive everywhere, and therefore can contain negative regions associated with non-classical features of the quantum state.

In Fig. 2, we show some representative examples. First, the vacuum state $|0\rangle$, represented by a symmetric Gaussian centered at the origin. Then, according to Eq. (2.19), a coherent state $|\alpha\rangle$ with $\alpha = 2$ corresponds to a displaced Gaussian. Also, an even Schrödinger cat state $|\mathcal{C}_+(\alpha)\rangle$ with $\alpha = 2$, defined later in Eq. (2.29), consists of displaying classical-like lobes at $\pm\alpha$ and a central region of quantum interference. Finally, the two-photon Fock state $|2\rangle$ is characterized by a rotationally symmetric ring structure, as a consequence of Eq. (2.14).

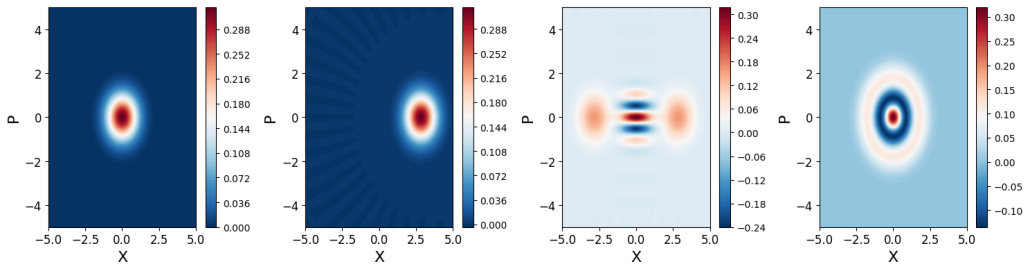


Figure 2: Wigner quasiprobability distributions in phase space for four elementary quantum states of the cavity field: (a) the vacuum state $|0\rangle$; (b) a coherent state $|\alpha = 2\rangle$; (c) an even Schrödinger cat state $|\mathcal{C}_+(\alpha = 2)\rangle$; and (d) the single-photon Fock state $|2\rangle$.

2.3 Photon-number parity

The photon-number parity is the central observable in the measurement protocol studied in this work. It distinguishes between states containing an even and an odd number of photons, so is defined as [10]

$$\hat{\Pi} = e^{i\pi\hat{n}} = (-1)^{\hat{n}}. \tag{2.23}$$

Since the Fock states are eigenstates of the photon-number operator, they are also eigenstates of the parity operator:

$$\hat{\Pi}\,|n\rangle = (-1)^n\,|n\rangle\,. \tag{2.24}$$

Thus, even-photon-number states have parity $+1$, whereas odd-photon-number states have parity -1.

The Hilbert space can therefore be decomposed into two orthogonal parity sectors,

$$\mathcal{H} = \mathcal{H}_{\text{even}} \oplus \mathcal{H}_{\text{odd}}. \tag{2.25}$$

The corresponding projectors are

$$\hat{P}_{\text{even}} = \frac{1}{2}\left(\hat{I} + \hat{\Pi}\right), \qquad \hat{P}_{\text{odd}} = \frac{1}{2}\left(\hat{I} - \hat{\Pi}\right) \tag{2.26}$$

and the probabilities of obtaining an even or odd outcome for a general state ρ in a projective parity measurement are therefore

$$p_{\text{even}} = \text{Tr}\left(\hat{P}_{\text{even}}\rho\right), \qquad p_{\text{odd}} = \text{Tr}\left(\hat{P}_{\text{odd}}\rho\right). \tag{2.27}$$

These projectors will play a central role in the following sections: they define the ideal projective measurements considered in the first part of the project and, later, provide the measurement operators used to describe continuous parity monitoring in the open quantum system.

In particular, the action of the parity operator on a coherent state follows directly from its expansion in the Fock basis (Eq. (2.14))

$$\hat{\Pi}\,|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}(-1)^n\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,. \tag{2.28}$$

Consequently, projecting a coherent state onto a definite parity sector selects only its even- or odd-photon-number components. The resulting states can be written as

$$|\mathcal{C}_\pm(\alpha)\rangle = \mathcal{N}_\pm\left(|\alpha\rangle \pm |-\alpha\rangle\right), \tag{2.29}$$

where $\mathcal{N}_\pm$ are normalization constants, and the signs $+$ and $-$ correspond to the even- and odd-parity sectors, respectively. The probability of obtaining each state is determined by the corresponding projective measurement probability in Eq. (2.27).

Thus, a projective parity measurement on a coherent state produces, depending on the measurement outcome, an even or odd Schrödinger cat state. These states are coherent superpositions of two states located at opposite points, α and $-\alpha$, in phase space. For sufficiently separated coherent states, their overlap becomes small, while their coherent superposition gives rise to interference features in phase space. The parity measurement therefore provides a simple mechanism through which a coherent, approximately classical-like state can be projected onto a non-classical quantum state.

An important property of the Wigner representation is its direct connection to the photon-number parity operator. In the convention used here, the Wigner function at a phase-space point α can be written as the expectation value of a displaced parity operator [2, 10], whose demonstration is developed in the appendix D

$$W(\alpha) = \frac{2}{\pi}\mathrm{Tr}\left[\rho\,\hat{D}(\alpha)\hat{\Pi}\hat{D}^{\dagger}(\alpha)\right] = \frac{2}{\pi}\left\langle \hat{D}(\alpha)\hat{\Pi}\hat{D}^{\dagger}(\alpha)\right\rangle. \tag{2.30}$$

Thus, the Wigner function can be interpreted in terms of a parity measurement performed after an appropriate displacement in phase space. This relation provides a direct connection between the main operations considered in this work and the phase-space representation of the cavity state. The sequence of displacement and parity measurements studied in the following section can therefore be interpreted simultaneously as a measurement-driven quantum dynamics and as a process that probes and modifies the phase-space structure of the cavity field.

3 Results and discussion

3.1 Projective parity measurements in a cavity QED system

The theoretical framework introduced above is now applied to an ideal measurement protocol which consists of the sequential application of a displacement in phase space and a projective measurement of photon-number parity. Starting from the vacuum state, a displacement $\hat{D}(\alpha)$ generates a coherent state (Eq. (2.18)). The parity measurement then projects the state onto either the even- or odd-parity subspace (Eq. (2.10)). The resulting state is subsequently displaced again, and the measurement is repeated. Since the outcome of each projective measurement is random, different sequences of outcomes generate different quantum trajectories. By repeating the simulation for a large number of independent realizations, ensemble-averaged quantities can also be obtained.

The protocol was simulated numerically using the `QuTiP` Python library [9] with the aim of investigating how repeated parity projections modify the state of the cavity and how the resulting dynamics can be characterized in phase space. In particular, we consider the parity probabilities, the mean field amplitude and the mean photon number, together with the Wigner function as a representation of the evolving quantum state.

Furthermore, the simulations were performed in a truncated Fock basis, as numerical calculations require a finite-dimensional Hilbert space. Since coherent states can contain contributions from large photon numbers, it is necessary to verify that the chosen cutoff is sufficiently large for the range of displacement amplitudes considered. We performed a convergence analysis by increasing N_{cut} and comparing the relevant observables, ensuring that the truncation does not introduce appreciable numerical errors in the simulations. The results bellow takes the $N_{\mathrm{cut}} = 30$.

For the first measurement, the parity expectation value can be obtained directly from the Fock-state expansion of the coherent state given in Eq. (2.14). Since parity acts on each Fock state according to Eq. (2.24), one obtains

$$\langle\hat{\Pi}\rangle_{\alpha} = \langle\alpha|\hat{\Pi}|\alpha\rangle = e^{-|\alpha|^2}\sum_{n=0}^{\infty}(-1)^n\frac{|\alpha|^{2n}}{n!} = e^{-|\alpha|^2}e^{-|\alpha|^2} = e^{-2|\alpha|^2}. \tag{3.1}$$

Thus, the parity expectation value is determined by the relative weight of the even- and odd-photon-number components of the coherent state. Using the projectors defined in Eq. (2.10), the probabilities of obtaining the two possible outcomes are

$$p_{\mathrm{even}} = \frac{1+\langle\hat{\Pi}\rangle_{\alpha}}{2} = \frac{1+e^{-2|\alpha|^2}}{2}, \qquad p_{\mathrm{odd}} = \frac{1-\langle\hat{\Pi}\rangle_{\alpha}}{2} = \frac{1-e^{-2|\alpha|^2}}{2}. \tag{3.2}$$

These expressions illustrate how the displacement amplitude controls the parity statistics. For small $|\alpha|$, the coherent state is dominated by the vacuum and low-photon-number components, with the even-photon-number sector therefore having a larger probability, as we can see in Fig. 4 (a). In contrast, for $|\alpha| \gg 1$, the exponential contribution becomes negligible and

$$p_{\mathrm{even}} \simeq p_{\mathrm{odd}} \simeq \frac{1}{2}. \tag{3.3}$$

The same behavior can be understood from the phase-space representation. As discussed in Ec. (2.29), a parity projection of $|\alpha\rangle$ produces an even or odd superposition of $|\alpha\rangle$ and $|-\alpha\rangle$. The

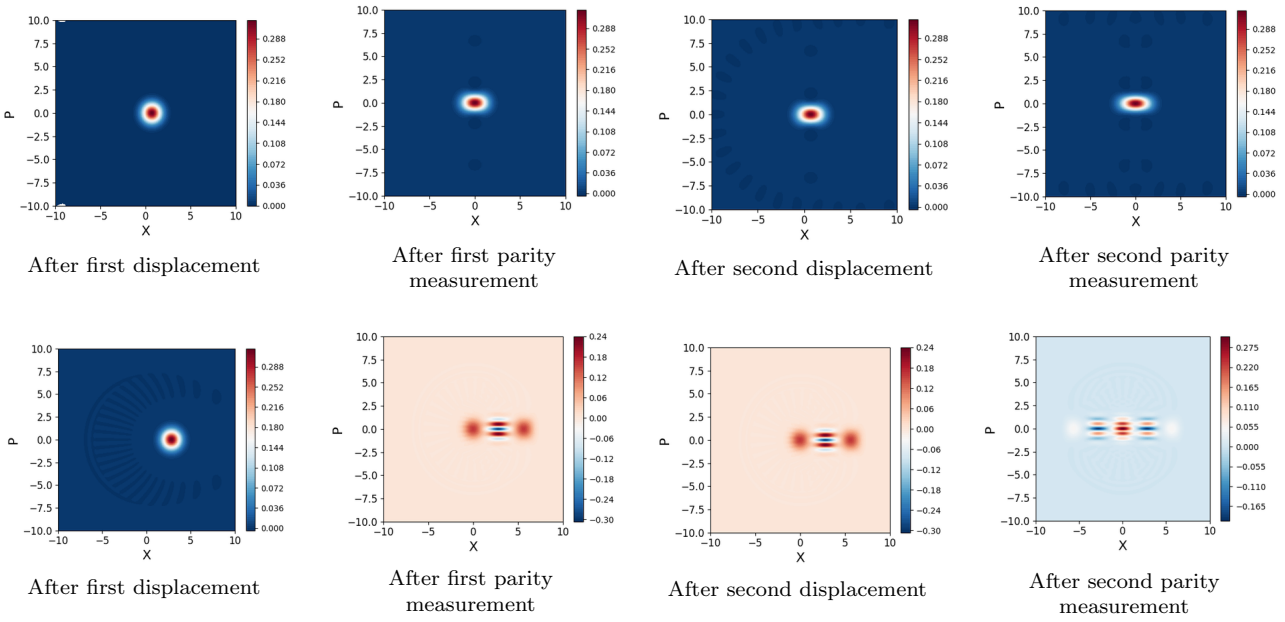


Figure 3: Wigner-function evolution under repeated displacement and projective parity measurements. The upper row corresponds to a small displacement ($\alpha = 0.5$), whereas the lower row corresponds to a larger displacement ($\alpha = 2$).

overlap between these two coherent states is

$$\langle \alpha | -\alpha \rangle = e^{-2|\alpha|^2}, \tag{3.4}$$

Therefore, for small displacements the two components strongly overlap, as can be seen in the top row of the Fig. 3, whereas for large displacements they become effectively distinguishable in phase space and the parity outcomes approach equal probabilities, as illustrated by the Wigner-function evolution in the bottom row of the Fig. 3.

After the first measurement, the situation changes because the state is no longer a coherent state but a parity-projected state. Consequently, the probabilities of the next parity outcome depends not only on the displacement amplitude but also on the result of the previous measurement.

This dependence can be illustrated analytically for a sufficiently large real displacement α, for which the overlap between $|\alpha\rangle$ and $|-\alpha\rangle$ can be neglected. If the first measurement yields the even outcome, the resulting state is approximately

$$|\psi_1\rangle \simeq \frac{1}{\sqrt{2}} \left(|\alpha\rangle + |-\alpha\rangle \right). \tag{3.5}$$

After applying the same displacement again, this state becomes

$$\hat{D}(\alpha) |\psi_1\rangle \simeq \frac{1}{\sqrt{2}} \left(|2\alpha\rangle + |0\rangle \right), \tag{3.6}$$

up to an irrelevant global phase.

The two components now have different parity content. The vacuum has purely even parity, while for sufficiently large $|\alpha|$ the coherent state $|2\alpha\rangle$ has approximately equal even and odd parity probabilities. Neglecting the small overlap between the two components, the probability of obtaining an even outcome in the second measurement is

$$p^{(2)}_{\text{even}} \simeq \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{3}{4}. \tag{3.7}$$

Thus, although the first parity measurement becomes approximately unbiased for large displacements, the subsequent measurements do not. Each measurement modifies the quantum state according to its outcome, while the following displacement redistributes its phase-space components. The probability of a given parity outcome therefore depends on the previous measurement history, giving rise to different quantum trajectories even when all realizations start from the same initial state and are subjected to the same displacement sequence. However, the ensemble average

in Fig. 3.20 reveals that, despite the stochastic nature of each individual quantum trajectory, the parity probabilities converge towards well-defined deterministic values. This behavior originates from the structure of the parity projectors and the geometry of the displaced states in phase space.

To characterize how the protocol modifies the energy of the field, we computed the average photon number $\langle n \rangle = \langle a^\dagger a \rangle$, after every displacement and parity measurement, averaging over a large ensemble of independent quantum trajectories.

From Eq. (2.19) we obtain

$$D^\dagger(\alpha)\, a^\dagger a\, D(\alpha) = (a^\dagger + \alpha^*)(a + \alpha) = a^\dagger a + \alpha a^\dagger + \alpha^* a + |\alpha|^2. \tag{3.8}$$

Taking the expectation value gives

$$\langle n \rangle' = \langle n \rangle + \alpha \langle a^\dagger \rangle + \alpha^* \langle a \rangle + |\alpha|^2. \tag{3.9}$$

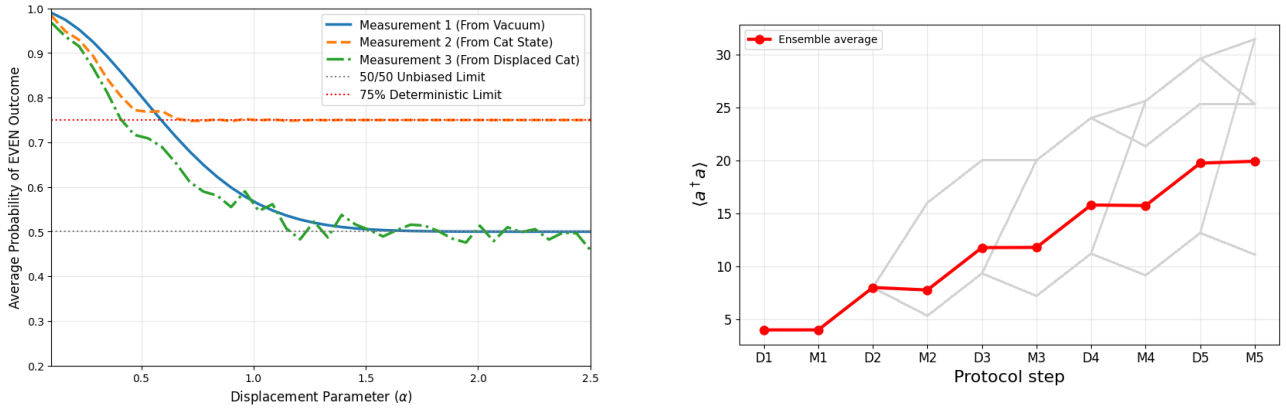
After every parity measurement the state is projected onto an eigenstate of the parity operator, for which $\langle a \rangle = \langle a^\dagger \rangle = 0$. Therefore,$\langle n \rangle' = \langle n \rangle + |\alpha|^2$.

For the simulations presented here, where $\alpha = 2$, every displacement injects exactly$|\alpha|^2 = 4$ photons on average, producing the sequence 4, 8, 12, 16, 20, which is in agreement with the numerical results in Fig. 4 (b).

Unlike the displacement operation, the parity measurement does not modify the average energy of the field. This follows from the fact that the photon-number operator commutes with the parity operator,

$$[a^\dagger a, \Pi] = 0 \tag{3.10}$$

In fact, the measurement only redistributes the state within either the even or odd Fock subspace, without creating or destroying photons on average.



(a) (b)

Figure 4: (a) Average probability of obtaining an even parity outcome as a function of the displacement parameter α for the first three successive steps of the protocol. (b) Mean photon number $\langle \hat{n} \rangle$ after successive displacement and parity-measurement steps for $\alpha = 2$. The red curve shows the ensemble average over independent quantum trajectories, while the grey lines are trajectories illustrating individual realizations.

However, the individual quantum trajectorie shown in Fig. 4 (b) behave very differently. Each parity measurement randomly projects the system onto either the even- or odd-parity subspace, causing the photon number of a single realization to jump upward or downward depending on the measurement outcome. These stochastic jumps correspond to the quantum measurement back-action and are shown as the gray trajectories in the figure.

When averaging over the ensemble of trajectories, these positive and negative fluctuations cancel statistically, producing the deterministic staircase followed by the red curve in Fig. 4 (b). The small deviations observed after each measurement originate from the finite number of Monte Carlo trajectories used in the simulation and decrease as the ensemble size increases.

3.2 Continuous parity monitoring and open-system dynamics

While the projective-measurement scheme relies on a sequence of discrete, strong interactions, continuous monitoring of the cavity constitutes an entirely different physical setup that must be modeled within the framework of open quantum systems. Rather than describing the dynamics through individual projective measurements separated by unitary operations, the interaction with the environment is represented by a continuous weak measurement process [11].

The dynamics of an open quantum system can be described by the Lindblad master equation

$$\frac{d\rho}{dt} = -i[\hat{H}, \rho] + \sum_k \mathcal{D}[\hat{L}_k]\rho, \tag{3.11}$$

where

$$\mathcal{D}[\hat{L}]\rho = \hat{L}\rho\hat{L}^\dagger - \frac{1}{2}\left\{\hat{L}^\dagger\hat{L}, \rho\right\} \tag{3.12}$$

is the Lindblad dissipator. Here, $\hat{H}$ describes the coherent evolution of the cavity, while the operators $\hat{L}_k$ describe the coupling to the environment.

For the present system, the environment is assumed to continuously monitor the photon-number parity. The two possible measurement outcomes correspond to the even- and odd-parity subspaces introduced previously. The corresponding collapse operators are therefore chosen as

$$\hat{L}_{\text{even}} = \sqrt{\gamma}\,\hat{P}_{\text{even}}, \qquad \hat{L}_{\text{odd}} = \sqrt{\gamma}\,\hat{P}_{\text{odd}}, \tag{3.13}$$

where γ is the measurement rate, so that larger values of γ lead to more frequent measurement events, whereas smaller values approach the weak-monitoring regime. The projectors are those defined in Eq. (2.26). Because the parity operator commutes with the photon-number operator, Ec. (3.10), this continuous monitoring acts as a quantum non-demolition (QND) [14] measurement that distinguishes between the two parity sectors while strictly conserving the photon-number distribution of the cavity field.

In fact, substituting Eq. (3.13) into the Lindblad equation gives the unconditional evolution (Appendix C)

$$\frac{d\rho}{dt} = -i[\hat{H}, \rho] + \frac{\gamma}{2}\left(\hat{\Pi}\rho\hat{\Pi} - \rho\right). \tag{3.14}$$

This equation describes a purely measurement-induced decoherence process. In particular, since $\hat{\Pi}^2 = \hat{I}$, it does not systematically change the populations of the two parity subspaces or exchange energy with the cavity. Instead, the continuous interaction with the environment progressively removes coherences between elements of ρ of different parity.

Although Eq. (3.14) provides a deterministic description of the density matrix, the same evolution can be viewed as the ensemble average of stochastic quantum trajectories. This alternative description is known as a quantum-jump unraveling of the master equation.

In this regard, the state of the cavity is conditioned on a particular measurement record and is denoted by $\rho_c(t)$ or $|\psi_c(t)\rangle$. During a small time interval dt, a detection event in channel k occurs with probability

$$p_k = dt\,\langle\hat{L}_k^\dagger\hat{L}_k\rangle_c, \tag{3.15}$$

where the subscript c indicates that the expectation value is evaluated for the current trajectory. When a jump occurs, the state is updated according to the corresponding collapse operator, concretely the jumps correspond to stochastic projections onto the even- and odd-parity sectors

$$|\psi_c\rangle \longrightarrow \frac{\hat{L}_k\,|\psi_c\rangle}{\sqrt{\langle\hat{L}_k^\dagger\hat{L}_k\rangle_c}}. \tag{3.16}$$

Between detection events, the state evolves continuously according to an effective Hamiltonian,

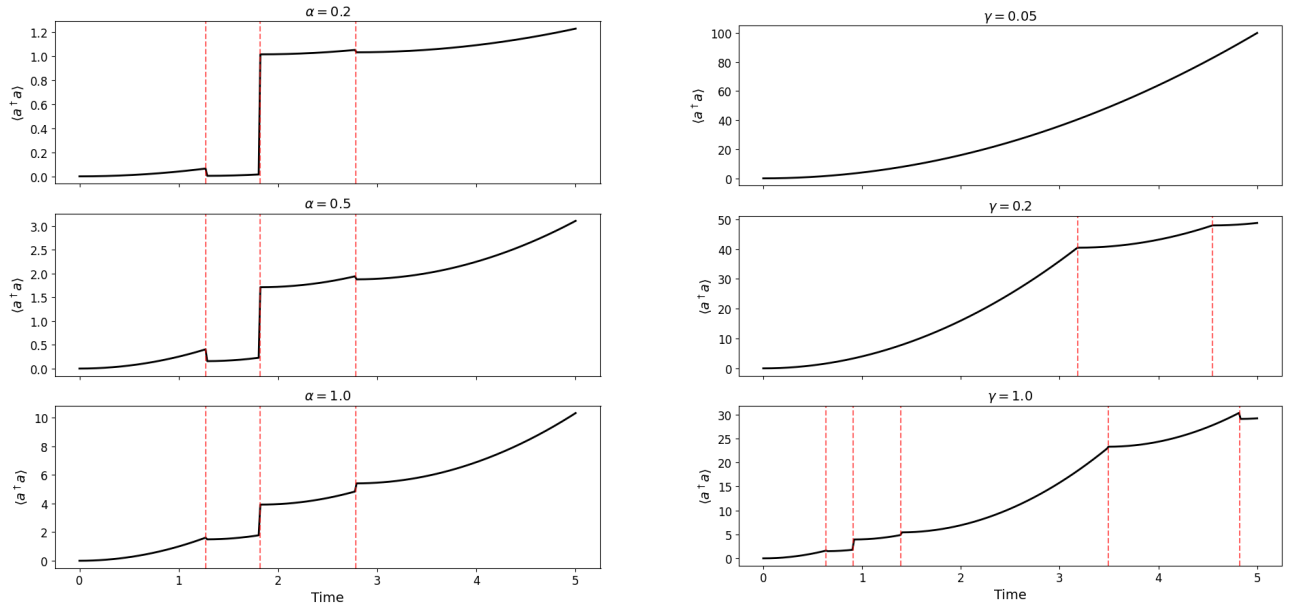
$$\hat{H} = i\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right), \tag{3.17}$$

where the parameter α therefore determines the strength of the coherent driving.

In the numerical simulations, the stochastic dynamics is obtained using the Monte Carlo wavefunction method implemented in QuTiP through `mcsolve`. Each run produces one possible realization of the monitored cavity dynamics. Since the jump times and measurement outcomes are stochastic, independent runs starting from the same initial state generally produce different trajectories, as demonstrated in Fig. 5.

To characterize individual trajectories, we monitor the mean photon number, since this observable provides a direct measure of the energy stored in the cavity and makes the stochastic effects of the measurement particularly visible

$$\langle \hat{n} \rangle_c = \langle \psi_c(t) | \hat{a}^\dagger \hat{a} | \psi_c(t) \rangle. \tag{3.18}$$



(a) Varying coherent driving α ($\gamma = 0.5$)

(b) Varying measurement rate γ ($\alpha = 2.0$)

Figure 5: Stochastic quantum trajectories of the cavity mean photon number $\langle \hat{n} \rangle_c$ under continuous parity monitoring. (a) Evolution under different coherent driving strengths α for a fixed measurement rate $\gamma = 0.5$. (b) Evolution for different measurement rates γ at a constant driving strength $\alpha = 2.0$.

The stochastic description becomes equivalent to the deterministic Lindblad description only after averaging over an ensemble of independent trajectories (Appendix A,B). If the state associated with trajectory i is

$$\rho_c^{(i)}(t) = |\psi_c^{(i)}(t)\rangle\langle\psi_c^{(i)}(t)|, \tag{3.19}$$

the unconditional density matrix is recovered as

$$\rho(t) = \mathbb{E}\left[\rho_c(t)\right] \simeq \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \rho_c^{(i)}(t), \tag{3.20}$$

where N_{traj} is the number of independent trajectories.

Consequently, for any observable $\hat{A}$,

$$\langle \hat{A} \rangle = \text{Tr}\left[\rho(t)\hat{A}\right] = \mathbb{E}\left[\langle \hat{A} \rangle_c\right]. \tag{3.21}$$

In particular, the photon-number dynamics obtained from the Monte Carlo simulation must converge to the solution of the Lindblad master equation when the number of trajectories is sufficiently large.

This correspondence is verified numerically in Fig. 6 by comparing the photon-number expectation value obtained from `mcsolve` with the deterministic solution obtained from `mesolve`. However, for a finite number of trajectories, the Monte Carlo result exhibits statistical fluctuations around the deterministic solution. As the number of trajectories is increased, these fluctuations become progressively smaller and the stochastic average converges to the Lindblad prediction.

The same correspondence can also be examined in phase space through the Wigner function. The numerical results in Fig. 7 show that the phase-space representation obtained from the ensemble of Monte Carlo trajectories agrees with the corresponding Lindblad evolution.

Besides, we can see that, for a single trajectory, the state is conditioned on one particular sequence of parity jumps , so different trajectories follow different random sequences, leading to slightly different final states. Nevertheless, when they are averaged, this spread produces a broader Wigner function and the elongated tail seen in Fig. 7, that reflects the statistical mixture of different conditional states rather than a broadening of a single coherent state.

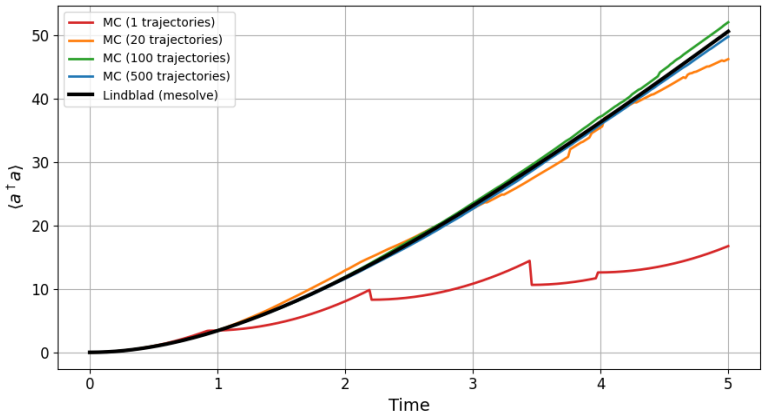


Figure 6: Convergence of the Monte Carlo wave-function ensemble towards the deterministic Lindblad master equation solution. The photon-number expectation value calculated from stochastic ensembles of increasing size ($N_{\text{traj}} = 1, 20, 100, 500$) is compared with the unconditional density matrix evolution.

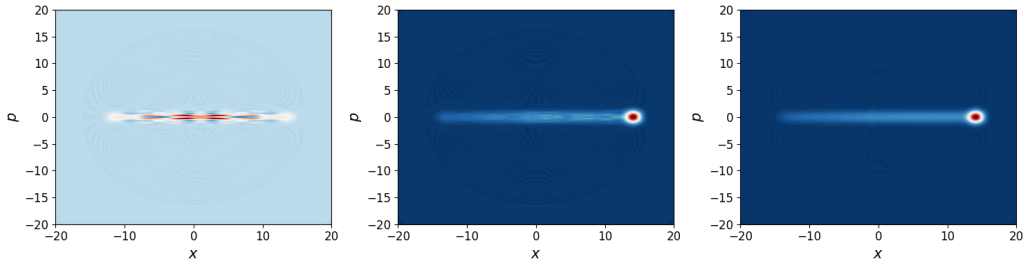


Figure 7: Wigner function of the cavity field under continuous parity monitoring. (Left) A single stochastic trajectory. (Center) The ensemble average over 500 independent trajectories. (Right) The unconditional state obtained via the Lindblad master equation.

4 Conclusion

This project has provided a first study of the dynamics of a cavity field under displacement and photon-number parity measurements, with particular emphasis on the transition from an ideal projective-measurement description to continuous monitoring in an open quantum system. Starting from the quantized description of a single cavity mode, the main concepts required to describe the system were introduced, including coherent states, phase-space representations, photon-number parity and the Wigner function.

The simulations of the projective measurement protocol showed how a parity measurement can transform an initially coherent state into an even or odd Schrödinger cat state. This provided a simple example of measurement-induced generation of non-classical states and showed how the overlap between the coherent components is directly related to the parity statistics and the appearance of interference structures in phase space. The Wigner representation was particularly useful for visualizing these effects and for connecting the displacement and parity operations with the evolution of the cavity state.

The study was then extended to continuous parity monitoring and open-system dynamics. The Lindblad master equation provides the deterministic description of the unconditional state, while the Monte Carlo wave-function method gives a stochastic description in terms of individual quantum trajectories. The numerical results confirmed that individual trajectories depend on the random sequence of measurement events, whereas their ensemble average converges to a Lindblad evolution. This correspondence was observed both in the photon-number dynamics and in the Wigner representation, providing a useful connection between the trajectory-level and ensemble descriptions of the same physical process.

Beyond the simulations presented here, the results of this study can serve as theoretical and numerical basis for a continuation of the project: the use of the cavity as a quantum reservoir [12, 13]. Quantum reservoir computing (QRC) exploits the natural dynamics of complex quantum systems to process temporal information, avoiding the need for fully programmable quantum circuits. In this framework, successive external inputs can be encoded through coherent displacements,

while the cavity dynamics, dissipation, and continuous parity monitoring provide the nonlinear and memory-dependent response required for reservoir computing. The present work therefore serves as a first characterization of the physical system and of the mechanisms that control its memory and decoherence, providing the necessary foundation for studying its computational capabilities as a quantum reservoir.

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Appendix

A Unconditional quantum state

The stochastic master equation describes the evolution of the *conditional* density matrix, $\rho_c(t)$, corresponding to a single realization of the continuous measurement process [11]. Since the sequence of quantum jumps is inherently random, each realization leads to a different quantum trajectory:

$$d\rho_c = dt\,\mathcal{L}\rho_c + \sum_k \Big(dN_k - dt\,\langle L_k^\dagger L_k\rangle_c\Big)\left(\frac{L_k\rho_c L_k^\dagger}{\langle L_k^\dagger L_k\rangle_c} - \rho_c\right) \tag{A.1}$$

where $\langle O\rangle_c = \mathrm{Tr}(O\rho_c)$, and the random variables dN_k describe the occurrence of quantum jumps in channel k during the infinitesimal time interval dt.

In case the individual measurement record is not accessible, or one is interested only in the average dynamics over many repetitions of the experiment, the relevant quantity is then the *unconditional* density matrix, defined as

$$\rho(t) = \mathbb{E}[\rho_c(t)]\,, \tag{A.2}$$

where $\mathbb{E}[\cdot]$ denotes the expectation value over all possible quantum trajectories. Equivalently, the unconditional density matrix corresponds to the deterministic solution of the Lindblad master equation,

$$\rho(t) = e^{\mathcal{L}t}\rho(0), \tag{A.3}$$

where $\mathcal{L}$ is the Liouvillian superoperator.

B Average jump probabilities

The probability that a quantum jump occurs in channel k during an infinitesimal time interval dt depends on the conditional state of the system and is given by

$$p_k^{(c)} = dt\,\mathrm{Tr}\Big(L_k^\dagger L_k\,\rho_c\Big)\,. \tag{B.1}$$

Since each quantum trajectory evolves according to a different sequence of measurement outcomes, the conditional density matrix ρ_c is generally different for every realization. The average jump probability obtained by averaging over all possible trajectories,

$$p_k = \mathbb{E}\Big[p_k^{(c)}\Big]\,. \tag{B.2}$$

Substituting the definition of the conditional probability:

$$p_k = \mathbb{E}\Big[dt\,\mathrm{Tr}\Big(L_k^\dagger L_k\,\rho_c\Big)\Big] \tag{B.3}$$

Since dt is a constant, it can be taken outside the expectation value, and the operator $L_k^\dagger L_k$ is fixed by the physical model and does not depend on the stochastic trajectory. Moreover, both the trace and the expectation value are linear operations. Therefore,

$$\mathbb{E}\Big[\mathrm{Tr}\Big(L_k^\dagger L_k\,\rho_c\Big)\Big] = \mathrm{Tr}\Big(L_k^\dagger L_k\,\mathbb{E}[\rho_c]\Big)\,. \tag{B.4}$$

Using the definition of the unconditional density matrix, $\mathbb{E}[\rho_c] = \rho$ one finally obtains

$$p_k = dt\,\mathrm{Tr}\Big(L_k^\dagger L_k\,\rho\Big)\,, \tag{B.5}$$

Overall, although every individual trajectory is characterized by a different sequence of quantum jumps, the average over infinitely many realizations defines a deterministic jump probability that depends only on the unconditional density matrix.

C Lindblad master equation for parity measurements

Starting from the stochastic equation,

$$\rho_c(t+dt) = \left(1 - \sum_k dN_k(t)\right)\frac{M_0\rho_c(t)M_0^\dagger}{1-\sum_k p_k^{(c)}(t)} + \sum_k dN_k(t)\frac{M_k\rho_c(t)M_k^\dagger}{p_k^{(c)}(t)} \tag{C.1}$$

one expects that averaging over all realizations of the stochastic variables $dN_k(t)$ should recover the unconditional quantum operation:

$$\rho(t+dt) = M_0\rho(t)M_0^\dagger + \sum_k M_k\rho(t)M_k^\dagger \tag{C.2}$$

However, this average cannot be performed by treating $dN_k(t)$ and $\rho_c(t)$ as statistically independent random variables. Indeed, the probability that a jump occurs is itself determined by the conditional state,

$$P(dN_k(t) = 1\,|\,\rho_c(t)) = p_k^{(c)}(t) \tag{C.3}$$

We make use of the law of total expectation,

$$\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}(X|Y)] \tag{C.4}$$

Let $g(\rho_c)$ be an arbitrary function, then,

$$\mathbb{E}[dN_k\,g(\rho_c)] = \mathbb{E}[\mathbb{E}\,(dN_k\,g(\rho_c)\,|\,\rho_c)]\,. \tag{C.5}$$

Since the function $g(\rho_c)$ is completely determined once the conditional state is fixed, it behaves as a constant inside the conditional expectation,

$$\mathbb{E}[dN_k\,g(\rho_c)] = \mathbb{E}[g(\rho_c)\,\mathbb{E}\,(dN_k\,|\,\rho_c)]\,. \tag{C.6}$$

The stochastic variable dN_k is a Bernoulli random variable that takes the values

$$dN_k = \begin{cases} 1, & \text{if a jump occurs in channel } k,\\ 0, & \text{otherwise.}\end{cases}$$

For a Bernoulli variable, the conditional expectation is equal to the probability of obtaining the value one.

$$\mathbb{E}\,(dN_k\,|\,\rho_c) = p_k^{(c)} \tag{C.7}$$

Replacing this result into the previous expression immediately yields the identity

$$\boxed{\mathbb{E}[dN_k(t)\,g(\rho_c(t))] = \mathbb{E}\Big[p_k^{(c)}(t)\,g(\rho_c(t))\Big]\,,} \tag{C.8}$$

This identity has a clear physical interpretation: although the occurrence of a quantum jump is random for each individual trajectory, once the conditional state is specified the jump probability is completely determined. Consequently, averaging over all trajectories replaces the stochastic indicator dN_k by its conditional mean value $p_k^{(c)}$.

The identity can now be applied to the jump contribution appearing in the stochastic update equation by choosing $g(\rho_c) = \frac{M_k \rho_c M_k^\dagger}{p_k^{(c)}}$.

Since the Kraus operators are fixed they can be taken outside the expectation value and using the definition of the unconditional density matrix,

$$\mathbb{E}\left[dN_k \frac{M_k \rho_c M_k^\dagger}{p_k^{(c)}}\right] = \mathbb{E}\left[p_k^{(c)} \frac{M_k \rho_c M_k^\dagger}{p_k^{(c)}}\right] = \mathbb{E}\left[M_k \rho_c M_k^\dagger\right] = M_k\, \mathbb{E}[\rho_c]\, M_k^\dagger = M_k \rho M_k^\dagger. \tag{C.9}$$

Therefore, averaging the stochastic evolution over all trajectories gives

$$\rho(t+dt) = M_0 \rho M_0^\dagger + \sum_k M_k \rho M_k^\dagger \tag{C.10}$$

which coincides with the unconditional quantum operation introduced previously. Expanding the Kraus operators,

$$M_0 = I - iH_{\text{eff}}dt, \qquad M_k = \sqrt{dt}\, L_k \tag{C.11}$$

to first order in dt finally recovers the Lindblad master equation,

$$\boxed{\frac{d\rho}{dt} = -i[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\left\{L_k^\dagger L_k, \rho\right\}\right).} \tag{C.12}$$

In the present work, the environment continuously monitors the parity of the cavity field. Instead of considering dissipative processes associated with energy loss, the interaction with the environment is described through two measurement channels corresponding to the even- and odd-parity subspaces, defined in Ec.(3.13). Substituting these collapse operators into the general Lindblad master equation gives

$$\begin{aligned}\frac{d\rho}{dt} =& -i[H,\rho] + \gamma\left(P_{\text{even}} \rho P_{\text{even}} - \frac{1}{2}\{P_{\text{even}}, \rho\}\right) \\ &+ \gamma\left(P_{\text{odd}} \rho P_{\text{odd}} - \frac{1}{2}\{P_{\text{odd}}, \rho\}\right).\end{aligned} \tag{C.13}$$

Using the completeness relation

$$P_{\text{even}} + P_{\text{odd}} = I, \tag{C.14}$$

the anticommutator terms simplify according to

$$-\frac{\gamma}{2}\left(\{P_{\text{even}}, \rho\} + \{P_{\text{odd}}, \rho\}\right) = -\gamma\rho. \tag{C.15}$$

The master equation can therefore be written in the compact form

$$\boxed{\frac{d\rho}{dt} = -i[H,\rho] + \gamma\left(P_{\text{even}} \rho P_{\text{even}} + P_{\text{odd}} \rho P_{\text{odd}} - \rho\right).} \tag{C.16}$$

Introducing the definitions of the parity projectors, Ec.(2.26), the two projector terms become

$$\begin{aligned}P_{\text{even}} \rho P_{\text{even}} &= \frac{1}{4}(I+\Pi)\rho(I+\Pi) \\ &= \frac{1}{4}\left(\rho + \rho\Pi + \Pi\rho + \Pi\rho\Pi\right),\end{aligned} \tag{C.17}$$

and

$$\begin{aligned}P_{\text{odd}} \rho P_{\text{odd}} &= \frac{1}{4}(I-\Pi)\rho(I-\Pi) \\ &= \frac{1}{4}\left(\rho - \rho\Pi - \Pi\rho + \Pi\rho\Pi\right).\end{aligned} \tag{C.18}$$

Adding both contributions, the cross terms cancel exactly,

$$P_{\text{even}}\rho P_{\text{even}} + P_{\text{odd}}\rho P_{\text{odd}} = \frac{1}{2}\left(\rho + \Pi\rho\Pi\right). \tag{C.19}$$

Substituting this result into the master equation yields

$$\begin{aligned}\frac{d\rho}{dt} &= -i[H,\rho] + \gamma\left[\frac{1}{2}(\rho + \Pi\rho\Pi) - \rho\right] \\ &= -i[H,\rho] + \frac{\gamma}{2}\left(\Pi\rho\Pi - \rho\right).\end{aligned} \tag{C.20}$$

Therefore, the unconditional dynamics can be written in the compact form

$$\boxed{\frac{d\rho}{dt} = -i[H,\rho] + \frac{\gamma}{2}\left(\Pi\rho\Pi - \rho\right).} \tag{C.21}$$

D Derivation of the Wigner Function as a Displaced Parity Expectation Value

In this appendix, we provide a detailed mathematical demonstration of the equivalence between the Wigner function at an arbitrary point α in phase space and the expectation value of the displaced photon-number parity operator, as stated in Eq. (2.30), following [10].

First, we evaluate the Wigner function at the origin of the phase space, $x = 0$ and $p = 0$. Substituting these values into Eq. (2.22)

$$W(0,0) = \frac{2}{\pi}\int_{-\infty}^{\infty} dy\, \langle -y|\rho|y\rangle. \tag{D.1}$$

By taking the trace of the product of the density operator ρ and the parity operator $\hat{\Pi}$, we obtain:

$$\begin{aligned}\text{Tr}[\rho\hat{\Pi}] &= \int_{-\infty}^{\infty} dx\, \langle x|\rho\hat{\Pi}|x\rangle \\ &= \int_{-\infty}^{\infty} dx\, \langle x|\rho|-x\rangle.\end{aligned} \tag{D.2}$$

Applying the change of variable $y = -x$ in the integral above, we can write:

$$\text{Tr}[\rho\hat{\Pi}] = \int_{-\infty}^{\infty} dy\, \langle -y|\rho|y\rangle. \tag{D.3}$$

Comparing Eq. (D.3) with Eq. (D.1), we establish the identity for zero displacement:

$$W(0) = \frac{2}{\pi}\text{Tr}[\rho\hat{\Pi}]. \tag{D.4}$$

To generalize this result for any point $\alpha = X_0 + iP_0$ in phase space, we consider the unitary displacement operator defined in Eq. (2.17). The action of the displacement $D(-\alpha)$ on the original field produces the translated state

$$\rho' = \hat{D}(-\alpha)\rho\hat{D}(\alpha). \tag{D.5}$$

We can understand the translation of the Wigner function by analyzing how the displacement operator acts on the quadrature eigenbasis $|x\rangle$. Since $\hat{P}$ acts as the translation generator for $\hat{X}$ under our commutation relation $[\hat{X},\hat{P}] = i/2$, the displacement operator can be shown to act as $\hat{D}(\alpha)|x\rangle = e^{2iP_0x + iX_0P_0}|x + X_0\rangle$. Applying this relation to the matrix elements of the translated state ρ', we obtain:

$$\langle X - y|\rho'|X + y\rangle = e^{4iP_0y}\langle (X + X_0) - y|\rho|(X + X_0) + y\rangle. \tag{D.6}$$

Inserting this expression into the definition of the Wigner function in Eq. (2.22), the phase factors combine to shift the arguments of the original Wigner function:

$$W_{\rho'}(X,P) = W_\rho(X + X_0, P + P_0). \tag{D.7}$$

Evaluating this translated Wigner function at the origin ($X = 0, P = 0$) directly yields the original Wigner function evaluated at the point (X_0, P_0), which corresponds to the complex amplitude α:

$$W_{\rho'}(0,0) = W_\rho(X_0, P_0) \equiv W(\alpha). \tag{D.8}$$

Using the relation derived in Eq. (D.4) for zero displacement, we can express the value of the shifted Wigner function at the origin as

$$W_{\rho'}(0,0) = \frac{2}{\pi}\mathrm{Tr}[\rho'\hat{\Pi}]. \tag{D.9}$$

Substituting the definition of the translated state ρ' from Eq. (D.5) into Eq. (D.9), we find:

$$W(\alpha) = \frac{2}{\pi}\mathrm{Tr}\left[\hat{D}^\dagger(\alpha)\rho\hat{D}(\alpha)\hat{\Pi}\right]. \tag{D.10}$$

Finally, we can rearrange the operators inside the trace and applying $\langle\hat{O}\rangle = \mathrm{Tr}[\rho\hat{O}]$

$$W(\alpha) = \frac{2}{\pi}\mathrm{Tr}\left[\rho\hat{D}(\alpha)\hat{\Pi}\hat{D}^\dagger(\alpha)\right] = \frac{2}{\pi}\left\langle\hat{D}(\alpha)\hat{\Pi}\hat{D}^\dagger(\alpha)\right\rangle, \tag{D.11}$$

which concludes the derivation of Eq. (2.30).

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